Software Architectural Design

A hospital expert system typically consists of the following main modules:

- **Knowledge Base:** Stores medical facts, rules, and expert knowledge relevant to diagnosis and treatment.
- **Inference Engine:** Processes information from patients and staff using reasoning strategies (both forward and backward chaining) to provide diagnosis, recommendations, and alerts.
- **User Interface:** Allows healthcare providers, patients, and admin staff to interact with the system—entering data, retrieving recommendations, and managing records.
- **Explanation Module:** Provides human-readable explanations for diagnoses, building trust and transparency.
- **Knowledge Acquisition Module**: Updates and expands the medical knowledge base through healthcare experts and machine learning.
- **Integration Layer**: Connects with external systems like Electronic Health Records (EHR), laboratory information systems, and mobile tracking modules.
- **Notification & Follow-Up System**: Tracks health events and ensures follow-up actions such as appointments, tests, and daily symptom monitoring.

Hardware Architectural Design

- **User Devices:** Computers, tablets, and smartphones used by healthcare workers, physicians, and patients.
- **Servers:** Central servers hosting the expert system's logic, databases, and inference engines.
- **Storage:** Secure storage system for EHRs, knowledge base, and transaction data (could be cloud, local, or hybrid).
- **Networking Infrastructure:** Routers, switches, and firewalls providing connectivity and security between devices and servers.
- **Peripheral Devices:** Medical equipment interfaced for real-time diagnostics and data collection (optional, depending on integration).

Network Architectural Design

A typical hospital expert system network deploys:

- **Web Server:** Hosts the expert system application, accessible through secure web interfaces.
- **Database Server:** Stores patient data, medical histories, and system rules.
- **Admin Device Node:** For admin staff to manage users and system configurations.

- **Patient & Physician Node:** Connected devices (computers or mobile gadgets) accessing the system via browser or app, securely through TCP/IP.
- **Firewall & ISP:** Protects sensitive data and ensures controlled internet access.
- **Cloud/Remote Access Layer:** Enables remote access for patients to monitor symptoms or communicate with care providers.

Class diagram (UML)

Class: Hospital

- name
- address
- phoneNumber
- staffList
- patientList
- doctorList

Class: Patient

- patientID
- name
- age
- gender
- address
- phone
- medicalHistory
- symptoms
- appointments
- bookAppointment()
- viewDiagnosis()

Class: Doctor

- doctorID
- name
- specialization

- contactInfo
- appointments
- prescribeMedicine()
- diagnosePatient()

Class: Appointment
- appointmentID
- patientID
- doctorID
- date
- time
- status

Class: MedicalRecord
- recordID
- patientID
- diagnosis
- treatment
- date

Class: ExpertSystem
- knowledgeBase
- inferenceEngine
- validateSymptoms()
- recommendTreatment()
- notifyFollowUp()

Class: Medicine
- medicineID
- name
- supplier

- dosage

- prescriptionList

Relationships:

- Hospital aggregates Patient, Doctor, and Appointment.

- Patient is associated with multiple MedicalRecords and Appointments.

- Doctor manages multiple Appointments and prescribes Medicine.

- Appointment has association with both Patient and Doctor.

- ExpertSystem interacts with MedicalRecord, Patient, Doctor, and Medicine (inference).

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